FRACTAL-WAVE ALGEBRA

PAGE 5: FROM RECURSIVE MODES TO PHYSICAL OBSERVABLES

The preceding pages established the generative sequence

$$I \to S \to Lu \to 0' \to \Phi \to F(\Phi) \to R \to O$$

and introduced recursion, invariance and dynamic stability. The next step is to connect the internal structure of a recursive wave mode with quantities that can be measured experimentally.

The central requirement is that an internal FWA state must possess an observable projection. Let

$$\Phi = \Phi(A, \varphi, \lambda, \omega, \chi, Lu, \mathcal{T})$$

where A is amplitude, φ phase, λ characteristic spatial scale, ω temporal frequency, χ chirality or orientation, Lu the differentiation parameter, and $\mathcal{T}$ the topological organization.

The observable quantities are obtained through a measurement map

$$O = M[\Phi].$$

Therefore the physical meaning of the internal variables is determined by how they enter M.

FREQUENCY AS A DYNAMIC PROPERTY

For a stable recursive mode, temporal evolution can be represented locally as

$$\Phi(\mathbf{r}, t) = A(\mathbf{r})e^{i\varphi(\mathbf{r},t)}.$$

The instantaneous angular frequency is

$$\omega = \frac{\partial \varphi}{\partial t}.$$

Likewise, the local wave vector is

$$\mathbf{k} = \nabla \varphi.$$

The familiar wave relations therefore arise naturally from the phase structure:

$$\omega = \partial_t \varphi, \qquad \mathbf{k} = \nabla \varphi.$$

The wavelength is then

$$\lambda = \frac{2\pi}{|\mathbf{k}|}.$$

Within FWA, these quantities are not fundamental static attributes of an object. They are derivatives of the underlying wave phase.

THE RECURSIVE MODE

A stable mode can be represented by

$$\Phi_p(\mathbf{r}, t) = A_p(\mathbf{r}) e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t + \varphi_0)}.$$

The mode remains identifiable when the recursive transformation preserves its defining invariants:

$$F(\Phi_p) \sim \Phi_p.$$

The symbol $\sim$ indicates structural equivalence rather than numerical identity.

This distinction is essential. A physical mode may change amplitude, phase or local configuration while remaining the same mode under the appropriate equivalence relation. Thus:

identity = invariant structure

rather than

identity = unchanged coordinates

ENERGY AS A MODE PROPERTY

Once a stable frequency is defined, the quantum relation

$$E = \hbar\omega$$

provides a direct connection between wave dynamics and measurable energy.

In FWA, this relation can be interpreted as the observable energy associated with a stable temporal mode. The important conceptual shift is that energy is not introduced as an independent substance. It is associated with the temporal structure of the mode.

Similarly,

$$p = \hbar k$$

connects momentum with the spatial phase gradient. Together,

$$E = \hbar\omega, \qquad \mathbf{p} = \hbar\mathbf{k}$$

establish the bridge

$$\text{recursive wave structure} \to (\omega, \mathbf{k}) \to (E, \mathbf{p}).$$

DISPERSION

A physical theory must specify how frequency depends on spatial structure. Let

$$\omega = \omega(\mathbf{k}, \Phi, Lu).$$

For a simple linear wave, $\omega = vk$. For a massive relativistic excitation,

$$E^2 = p^2c^2 + m^2c^4.$$

Using $E = \hbar\omega$ and $p = \hbar k$, this becomes

$$\hbar^2\omega^2 = \hbar^2k^2c^2 + m^2c^4.$$

FWA can therefore treat the observed dispersion relation as an emergent property of the allowed recursive modes.

The stronger claim—that mass itself emerges from the recursive structure—requires an explicit derivation of the corresponding dispersion relation rather than simply identifying m with an FWA parameter.

MASS AS A STABLE RECURSIVE SCALE

If a stable mode possesses a characteristic internal frequency ω_0, then its rest energy can be represented as

$$E_0 = \hbar\omega_0.$$

For a massive object, $E_0 = mc^2$. Therefore,

$$mc^2 = \hbar\omega_0$$

and consequently

$$m = \frac{\hbar\omega_0}{c^2}.$$

This provides a possible FWA interpretation of mass:

mass $\leftrightarrow$ stable internal temporal mode

This is not yet a derivation of the Standard Model mass spectrum. It is a structural hypothesis that can become physically meaningful only if FWA derives the observed masses from its underlying parameters without inserting them as empirical inputs.

CHARACTERISTIC LENGTH

A stable mode can possess a characteristic length

$$\lambda_0 = \frac{2\pi c}{\omega_0}$$

for a relativistic wave moving at c.

Using $\omega_0 = \frac{mc^2}{\hbar}$, we obtain $\lambda_0 = \frac{2\pi\hbar}{mc}$. Therefore

$$\lambda_0 = \frac{h}{mc},$$

which is the Compton wavelength.

This demonstrates an important methodological point: familiar physical scales can be interpreted as characteristic lengths of stable wave modes, but the theory must still explain why a particular recursive mode selects the observed scale.

Lu AND PHYSICAL SCALE

The FWA differentiation parameter Lu can be introduced as a dimensionless structural parameter connecting an underlying reference scale to an emergent scale. A generic relation is

$$L_{\text{obs}} = \frac{L_*}{f(Lu)}$$

where L_* is a fundamental reference scale and $f(Lu)$ is determined by the recursive dynamics.

For the specific square-root scaling proposed within FWA,

$$\lambda = \frac{\lambda_P}{\sqrt{Lu}}$$

where λ_P is the Planck length.

This relation is potentially powerful because it converts a dimensionless structural parameter into a macroscopic or experimentally accessible length scale. However, it must be treated as a testable FWA relation, not as an established physical identity. Its significance depends on whether Lu can be independently derived and whether the resulting predictions agree with measurements across more than one system.

THE OBSERVABLE MAP

The complete transition from internal structure to measurement can now be written as

$$(I, S, Lu, 0', \Phi) \to F \to \Phi_p \to (\omega, \mathbf{k}) \to (E, \mathbf{p}) \to O.$$

This establishes the central physical bridge of the framework.

Information determines structural differentiation. Structural differentiation participates in recursive dynamics. Recursive dynamics generates stable modes. Stable modes possess measurable frequencies and spatial scales. Those quantities map to experimentally defined observables.

Thus the central FWA proposition becomes:

Observable physics = projection of stable recursive wave modes